



TECHNICAL DOCUMENT

Cybersecurity Incident Response Dashboard for SMEs

Company:

CyberShield Solutions

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INDEX

Table of Contents

I.	Introduction	Page 3
II.	Project Objectives	Page 4
III.	Problem Overview	Page 4
IV.	Technologies and Tools Used	Page 5
V.	Power BI Desktop (Local Dashboard)	Page 5
VI.	Excel Database	Page 6
VII.	Power BI Service	Page 7
VIII.	Data_Feeder System	Page 8
IX.	Python Automation File	Page 10
	a) Data Feeder	Page 10
	b) Alert Engine	Page 10
X.	Alert Engine	Page 10
XI.	Programming Language Used	Page 11
XII.	Cybersecurity Incident Indicators	Page 12
XIII.	System Workflow	Page 12
XIV.	System Architecture	Page 13
XV.	Advantages of the System	Page 14
XVI.	Future Improvements	Page 15
XVII.	Conclusion	Page 15

1. Introduction

Cybersecurity threats are continuously increasing, especially for Small and Medium-sized Enterprises (SMEs) that often lack advanced security infrastructure and monitoring systems. The project **Cybersecurity Incident Response Dashboard for SMEs** was developed to provide an automated and intelligent monitoring platform capable of collecting, processing, analyzing, and visualizing cybersecurity incident data.

The system combines Excel databases, Python automation scripts, Power BI Desktop dashboards, and Power BI Service cloud synchronization to create a complete cybersecurity monitoring environment.

This project helps organizations:

- Monitor cybersecurity incidents
- Analyze security threats
- Track incident trends
- Generate alerts
- Improve incident response processes

2. Project Objectives

The main objectives of the project are:

- Monitor cybersecurity incidents.
- Store and manage incident data efficiently.
- Automate data insertion and processing.
- Generate interactive dashboards for analysis.
- Enable automatic dashboard refreshing using Power BI Service.
- Trigger alerts when data increases.
- Provide insights for faster incident responses.

3. Project Overview

The project is designed as an automated cybersecurity monitoring system where cybersecurity-related data is inserted into an Excel database through a Python-based Data_Feeder system. The data is processed automatically and visualized using Power BI dashboards.

An alert engine continuously monitors incident thresholds. If abnormal activity or data increase is detected, automatic alerts are generated and logged.

The architecture allows SMEs to:

- Visualize security incidents
- Detect abnormal behavior
- Analyze attack patterns

- Improve security decision-making

4. Technologies and Tools Used

Technology	Purpose
Python	Automation and data processing
Microsoft Excel	Database storage
Power BI Desktop	Dashboard creation
Power BI Service	Cloud dashboard refresh
Data_Feeder	Automated data insertion
Alert Engine	Threshold monitoring and alert generation

5. Power BI Desktop (Local Dashboard)

Purpose

Power BI Desktop is used locally to design and create cybersecurity dashboards.

Explanation

The dashboard connects directly to the Excel database and displays:

- Cybersecurity incident statistics
- Threat severity levels
- Login attempt monitoring
- Malware activity
- DDoS attack indicators
- Security KPI indicators
- Incident response metrics

The dashboards include:

- Graphs
- Charts
- KPI cards
- Trend analysis visuals
- Incident summaries

Benefits

- Interactive visualization
- Easy analysis of incidents
- Centralized monitoring
- User-friendly interface

6. Excel Database

Purpose

Excel is used as the primary database for storing cybersecurity incident data.

Explanation

The Excel workbook contains multiple sheets including:

- Data Sheet
- Alert Log
- Configuration Sheet

Stored Data Includes

- Incident ID
- Incident Type
- Severity Level
- Timestamp
- Affected Systems
- Login Attempts
- Network Traffic
- IDS Alerts
- Suspicious Processes
- Alert Status

The database acts as a lightweight solution suitable for SMEs.

Benefits

- Easy integration with Power BI
- Simple maintenance
- Flexible storage structure
- Cost-effective database system

7. Power BI Service

Purpose

Power BI Service is used to publish dashboards online and refresh data automatically.

Explanation

After creating the dashboard in Power BI Desktop:

1. Reports are published to Power BI Service.
2. Scheduled refresh is configured.
3. Updated Excel data is synchronized automatically.

This enables dashboard updates whenever new data is inserted by the automation scripts.

Features

- Cloud accessibility
- Scheduled refresh
- Dashboard sharing
- Remote monitoring

Benefits

- Automatic updates
- Centralized access
- Improved collaboration
- Continuous monitoring

8. Data_Feeder System

Purpose

The Data_Feeder automatically inserts cybersecurity monitoring data into the Excel database.

Explanation

The data_feeder.py script generates and injects realistic cybersecurity incident data.

Supported Simulations

- DDoS attacks
- Phishing attacks
- Brute force attacks
- Malware incidents

The feeder automates:

- Data insertion
- Data formatting
- Database updates
- Incident simulation

Example Commands

Insert normal Data

```
python data_feeder.py
```

Insert 24 Hours of Data

```
python data_feeder.py --rows 24
```

Simulate DDoS Attack

```
python data_feeder.py --rows 3 --incident ddos
```

Simulate Phishing Attack

```
python data_feeder.py --rows 3 --incident phishing
```

Benefits

- Reduces manual data entry
- Improves testing capability
- Generates realistic datasets
- Supports cybersecurity simulations

9. Python Automation File

Purpose

Python automation files are used to automate system operations and cybersecurity monitoring tasks.

Main Automation Components

1. Data Feeder

Responsible for generating and inserting incident data.

2. Alert Engine

Responsible for:

- Monitoring thresholds
- Detecting abnormal increases
- Sending alerts
- Logging incidents

10. Alert Engine

Purpose

The alert engine monitors cybersecurity metrics and generates alerts when threshold values are exceeded.

Workflow

1. Read the latest data from Excel.
2. Compare values against threshold configurations.
3. Detect abnormal increases.
4. Trigger alert notifications.
5. Store alert logs in Excel.

Features

- Automatic alert generation
- Threshold monitoring
- Incident logging
- Continuous monitoring

11. Programming Language Used

Python

Explanation

Python is the main programming language used throughout the project.

It is used for:

- Automation
- Data handling

- Incident processing
- Alert generation
- Excel integration
- Monitoring operations

Python Libraries Used

- openpyxl
- schedule
- os
- datetime

Advantages of Python

- Easy syntax
- Strong automation support
- Excellent file handling
- Fast development
- Large cybersecurity ecosystem

12. Cybersecurity Incident Indicators

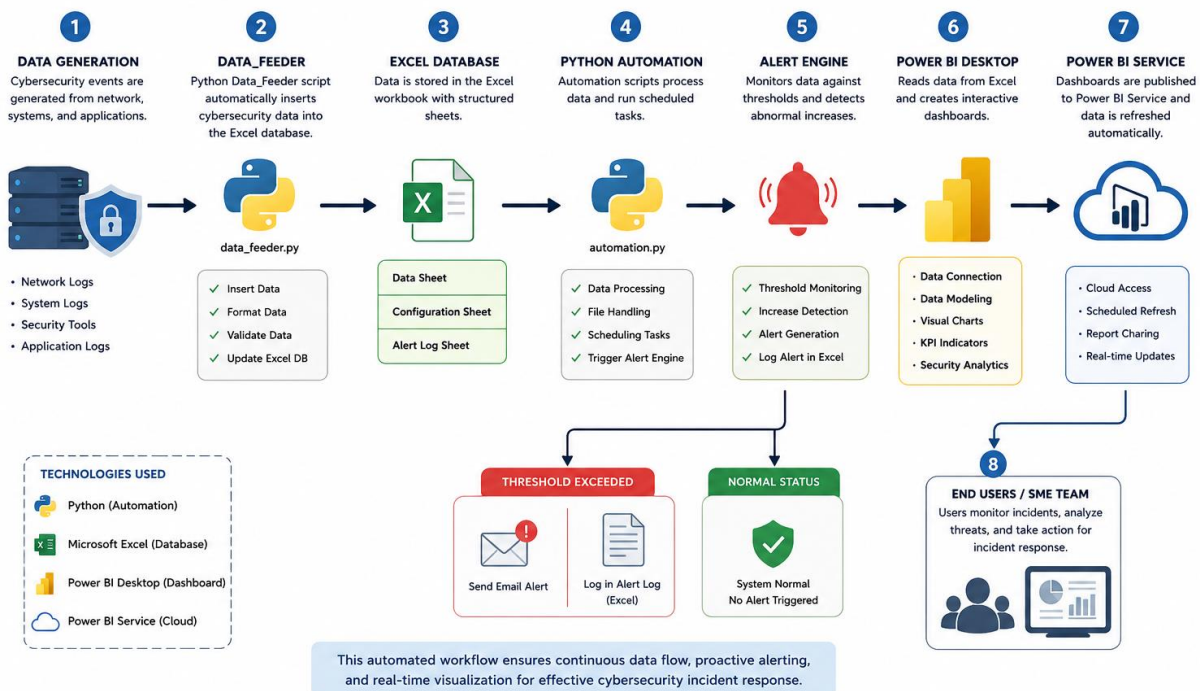
Incident Type	Indicators
DDoS Attack	High traffic, refused connections
Phishing	High email activity, suspicious logins
Brute Force	Multiple login attempts
Malware	Suspicious processes, unusual traffic

13. System Workflow

Step-by-Step Process

1. Cybersecurity data is generated.
2. Data_Feeder inserts records into Excel.
3. Python automation processes data.
4. Alert engine checks thresholds.
5. Alerts are generated if abnormal activity is detected.
6. Power BI Desktop reads the Excel database.
7. Dashboards are refreshed in Power BI Service.
8. Users monitor incidents through dashboards.

CYBERSECURITY INCIDENT RESPONSE DASHBOARD FOR SMEs SYSTEM WORKFLOW



14. System Architecture

Components

- Excel Database
- Data_Feeder Script
- Alert Engine
- Power BI Desktop
- Power BI Service

Architecture Flow

Cybersecurity Data



Data_Feeder (Python)



Excel Database



Alert Engine



Power BI Desktop



Power BI Service



Interactive Dashboard

15. Advantages of the System

- Automated cybersecurity monitoring
- Interactive dashboards
- Automatic alert generation
- Lightweight database solution
- Cloud synchronization
- Improved incident visibility
- Cost-effective for SMEs

16. Future Improvements

Possible future enhancements include:

- Integration with SQL databases
- Real-time SIEM integration
- AI-based threat prediction
- SMS alert notifications
- Web-based dashboard portal
- Advanced threat analytics

17. Conclusion

The **Cybersecurity Incident Response Dashboard for SMEs** project successfully demonstrates how automation, visualization, and data management technologies can be combined to create an efficient cybersecurity monitoring solution.

By integrating:

- Python automation
- Excel database management
- Power BI visualization
- Automated alert systems

The system provides SMEs with a powerful and affordable platform for monitoring cybersecurity incidents, analyzing threats, and improving incident response capabilities.

The project reduces manual effort, improves operational efficiency, and enables organizations to make better cybersecurity decisions through data-driven insights.